E-commerce Performance Analysis

1.Project Overview

This project analyzes an e-commerce dataset to understand operational performance and customer behavior. Using SQL for data processing and Tableau for visualization, the analysis focuses on key areas such as delivery efficiency, customer growth, and service quality. Two interactive dashboards were created to highlight trends in order fulfillment, customer engagement, and the impact of delivery delays on customer satisfaction. The goal is to derive actionable insights that can help improve logistics performance and enhance customer retention.

2. Dataset Summary

The analysis uses the Olist Brazilian E-commerce dataset, which contains real transaction data from an online marketplace.

- Covers ~100,000 orders from 2016 to 2018
- Includes multiple related tables such as:
    - Orders (timestamps, status)
    - Customers (location data)
    - Order items (products, sellers)
    - Payments (order value)
    - Reviews (customer feedback)

3. Exploratory Data Analysis using Python

Exploratory Data Analysis (EDA) was performed in Python using **Pandas** to understand data structure, quality, and initial patterns before building SQL views and dashboards.

1. Data Loading

All CSV files were loaded into DataFrames.

2. Initial Exploration

Checked structure and sample data.

```
[2]: import pandas as pd
```

```
[15]: customers=pd.read_csv("/Users/sneha_janu18/Downloads/olist_customers_dataset.csv")
customers.head()
```

[15]:

	customer_id	customer_unique_id	customer_zip_code_prefix	customer_city	customer_state
0	06b8999e2fba1a1fbc88172c00ba8bc7	861eff4711a542e4b93843c6dd7febb0	14409	franca	SP
1	18955e83d337fd6b2def6b18a428ac77	290c77bc529b7ac935b93aa66c333dc3	9790	sao bernardo do campo	SP
2	4e7b3e00288586ebd08712fdd0374a03	060e732b5b29e8181a18229c7b0b2b5e	1151	sao paulo	SP
3	b2b6027bc5c5109e529d4dc6358b12c3	259dac757896d24d7702b9acbbff3f3c	8775	mogi das cruzes	SP
4	4f2d8ab171c80ec8364f7c12e35b23ad	345ecd01c38d18a9036ed96c73b8d066	13056	campinas	SP

```
[4]: payments=pd.read_csv("/Users/sneha_janu18/Downloads/olist_order_payments_dataset.csv")
payments.head()
```

[4]:

	order_id	payment_sequential	payment_type	payment_installments	payment_value
0	b81ef226f3fe1789b1e8b2acac839d17	1	credit_card	8	99.33
1	a9810da82917af2d9aefd1278f1dcfa0	1	credit_card	1	24.39
2	25e8ea4e93396b6fa0d3dd708e76c1bd	1	credit_card	1	65.71
3	ba78997921bbcdc1373bb41e913ab953	1	credit_card	8	107.78
4	42fdf880ba16b47b59251dd489d4441a	1	credit_card	2	128.45

```
[5]: geolocation=pd.read_csv("/Users/sneha_janu18/Downloads/olist_geolocation_dataset.csv")
geolocation.head()
```

3. Missing Value Analysis

```
[6]: orders.isnull().sum()
```

```
[6]: order_id                          0
     customer_id                       0
     order_status                      0
     order_purchase_timestamp          0
     order_approved_at               160
     order_delivered_carrier_date   1783
     order_delivered_customer_date  2965
     order_estimated_delivery_date     0
     dtype: int64
```

```
[9]: products.isnull().sum()
```

```
[9]: product_id                     0
     product_category_name        610
     product_name_lenght          610
     product_description_lenght   610
     product_photos_qty           610
     product_weight_g               2
     product_length_cm              2
     product_height_cm              2
     product_width_cm               2
     dtype: int64
```

4. Database Integration

```
Note: you may need to restart the kernel to use updated packages.
```

```
[25]: customers.to_sql("customers", engine, if_exists="replace", index=False)
payments.to_sql("order_payments", engine, if_exists="replace", index=False)
orders.to_sql("orders", engine, if_exists="replace", index=False)
order_items.to_sql("order_items", engine, if_exists="replace", index=False)
reviews.to_sql("order_reviews", engine, if_exists="replace", index=False)
products.to_sql("products", engine, if_exists="replace", index=False)
sellers.to_sql("sellers", engine, if_exists="replace", index=False)
geolocation.to_sql("geolocation", engine, if_exists="replace", index=False)
product_category.to_sql("product_category",engine,if_exists="replace",index=False)

print("Tables successfully loaded into PostgreSQL.")
```

```
Tables successfully loaded into PostgreSQL.
```

4. Data Analysis Using SQL

SQL was used to perform structured analysis on the relational dataset in PostgreSQL by joining multiple tables and creating aggregated insights.

1. Monthly Revenue Trend- Tracks business growth over time.

	month timestamp without time zone	revenue numeric
1	2017-07-01 00:00:00	566403.93
2	2018-05-01 00:00:00	1128836.69
3	2018-07-01 00:00:00	1027903.86
4	2016-10-01 00:00:00	46566.71
5	2018-08-01 00:00:00	985414.28
6	2018-02-01 00:00:00	966510.88
7	2018-06-01 00:00:00	1012090.68

2. Yearly Revenue Growth-Measures year-over-year business performance.

	years timestamp without time zone	revenue numeric	revenue_growth numeric
1	2016-01-01 00:00:00	46586.33	0
2	2017-01-01 00:00:00	6922900.24	14760.37
3	2018-01-01 00:00:00	8452975.20	22.10

3. Top 10 Product Categories by Revenue-

	product_category_name text	value numeric
1	beleza_saude	1412089.53
2	relogios_presentes	1264333.12
3	cama_mesa_banho	1225209.26
4	esporte_lazer	1118256.91
5	informatica_acessorios	1032723.77
6	moveis_decoracao	880329.92
7	utilidades_domesticas	758392.25
8	cool_stuff	691680.89
9	automotivo	669454.75
10	ferramentas_jardim	567145.68

4. Monthly Unique Customers

	month timestamp without time zone	unique_customer bigint
1	2016-09-01 00:00:00	1
2	2016-10-01 00:00:00	265
3	2016-12-01 00:00:00	1
4	2017-01-01 00:00:00	750
5	2017-02-01 00:00:00	1653
6	2017-03-01 00:00:00	2546
7	2017-04-01 00:00:00	2303
8	2017-05-01 00:00:00	3546
9	2017-06-01 00:00:00	3135
10	2017-07-01 00:00:00	3872
11	2017-08-01 00:00:00	4193
12	2017-09-01 00:00:00	4150

5. Repeat Purchase Rate

	repeat_purchase_rate numeric
1	3.00027849782557467000

6. Delivery Delay Percentage

	percentage_delay numeric
1	8.11169385766703289900

8. State-wise Delivery Time

	state text	delivery_time interval
1	RR	28 days 33:18:03.97561
2	AP	26 days 28:26:29.8507...
3	AM	25 days 34:13:25.6137...
4	AL	24 days 13:03:09.1032...
5	PA	23 days 18:33:00.0211...
6	MA	21 days 13:45:05.1673...
7	SE	21 days 12:28:29.7074...

5. Dashboard in Tableau

Tableau was used to create interactive dashboards for visualizing key business metrics and insights derived from SQL analysis.

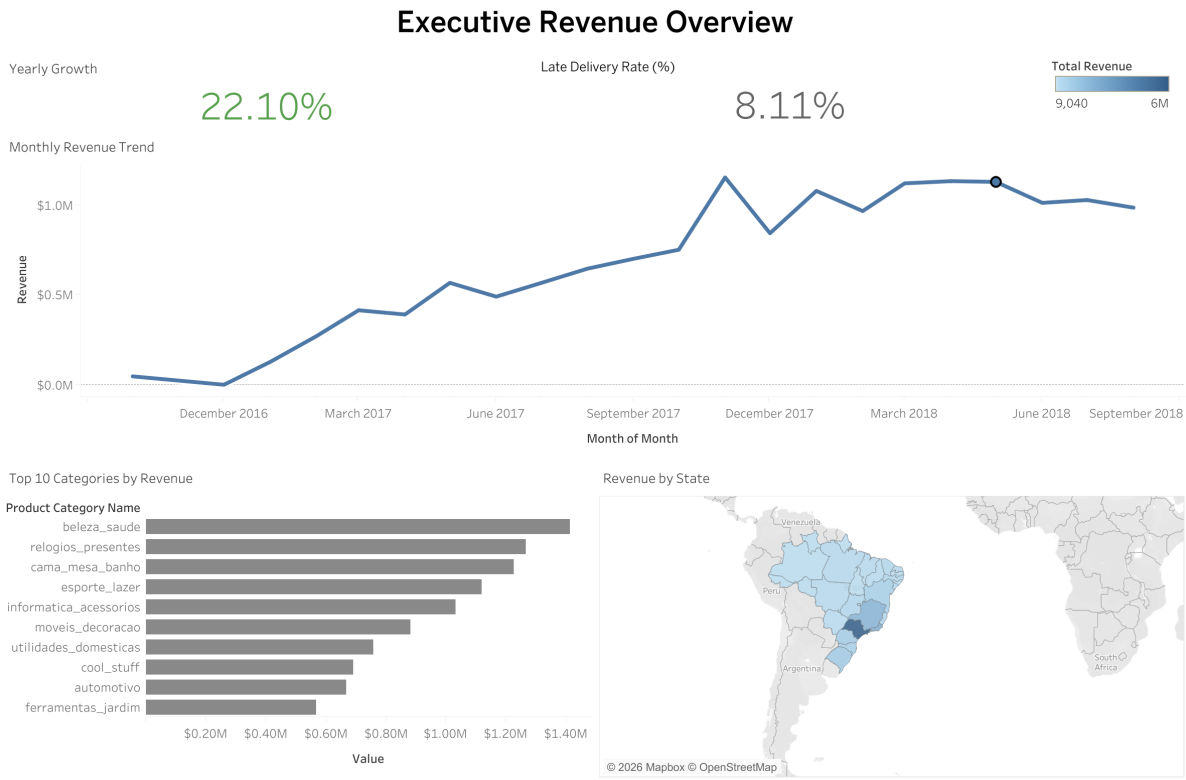
- Convert SQL outputs into visual insights
- Enable quick decision-making
- Present data in an intuitive and structured format

Dashboard 1 —Executive Revenue Overview

Objective: Analyze operational performance and delivery efficiency.

Visuals Included:

- **Monthly Revenue Trend (Line Chart)**
  Shows revenue growth over time
- **Yearly Growth (KPI)**
  Highlights overall business growth
- **Top Categories by Revenue (Bar Chart)**
  Identifies major revenue drivers
- **Revenue by State (Map)**
  Displays geographic contribution
- **Delayed Delivery % (KPI)**
  Measures delivery performance

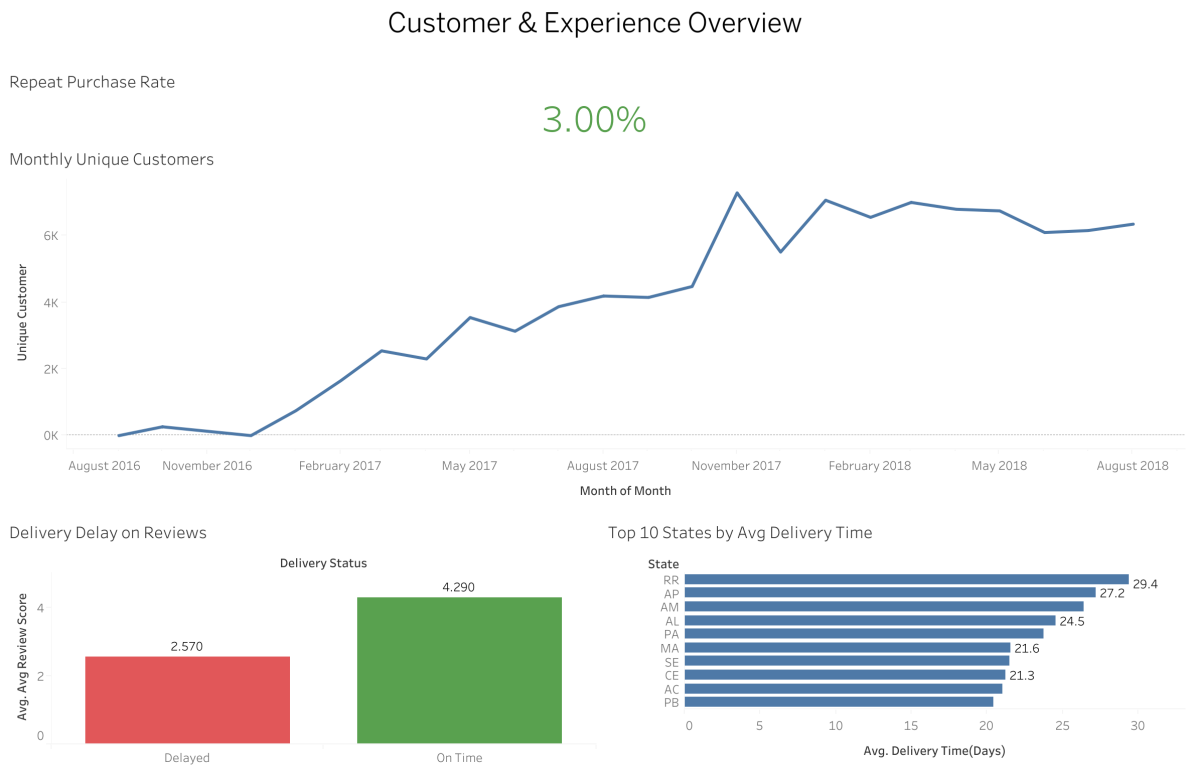


Dashboard 2 — Customer & Experience Overview

Objective: Evaluate customer behavior and service quality.

Visuals Included:

- **Monthly Unique Customers (Line Chart)**
  Tracks customer growth and engagement
- **Repeat Purchase Rate (KPI)**
  Measures customer retention (~3%)
- **Impact of Delivery Delay on Reviews (Bar Chart)**
  Compares satisfaction between delayed and on-time orders
- **Top 10 States by Delivery Time (Bar Chart)**
  Identifies regions with delivery inefficiencies



6. Business Recommendations

- Improve delivery in slow states to reduce delays and low reviews
- Increase retention with offers and loyalty programs (repeat rate is low)
- Fix high-cancellation categories (inventory, pricing, delivery issues)
- Scale top-performing categories with marketing and stock
- Remove or control low-rated sellers to protect quality
- Improve customer communication for better experience and ratings